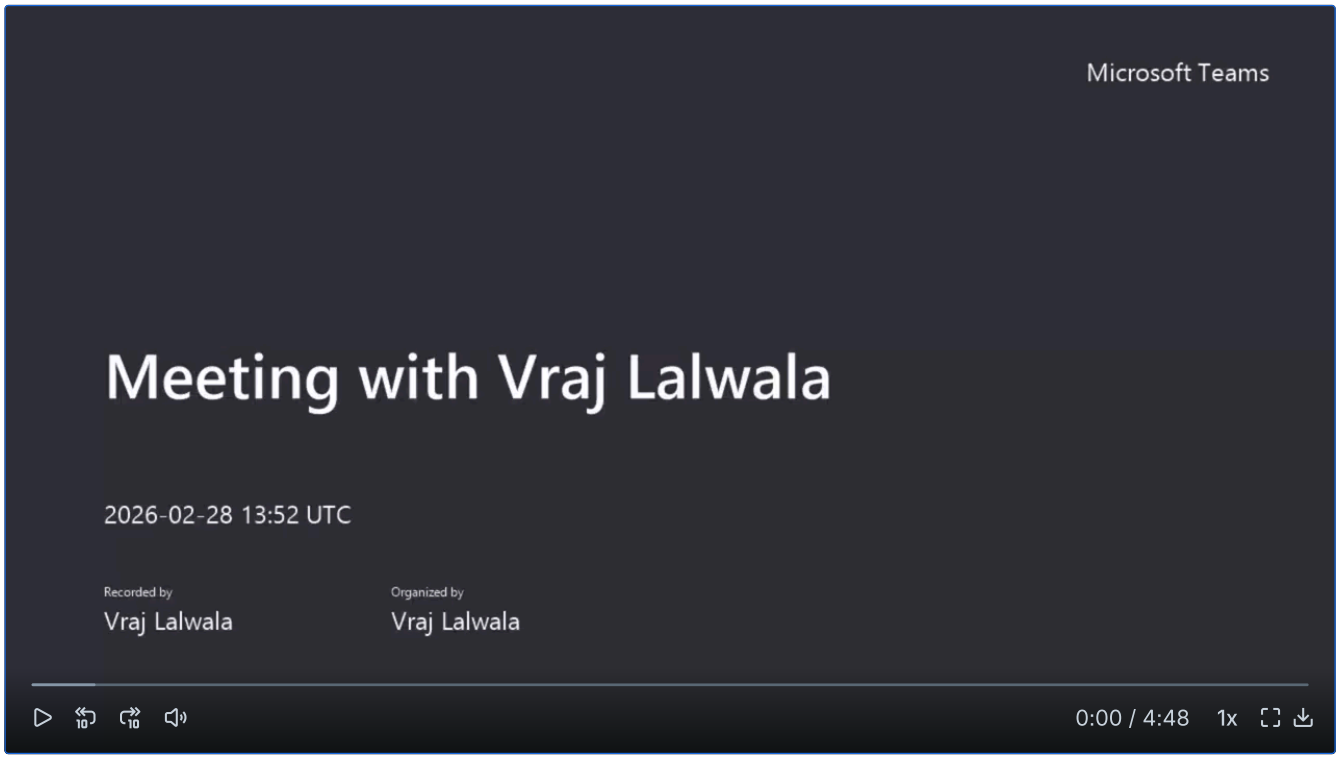
Difference Between Containerization in AWS Lambda and AWS ECS



1. Introduction

Modern cloud-native applications increasingly use **containerization** to package applications along with their dependencies into portable and consistent runtime environments.

In this project, the temp directory monitoring collector was containerized and deployed using two AWS compute services:

- **AWS Lambda (Container Image Based Deployment)**
- **AWS ECS Fargate (Container-Based Execution)**

Although both services run containers, their execution models, scaling mechanisms, and operational responsibilities differ significantly.

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2. Purpose of Containerization in This Task

The monitoring collector application performs the following operations:

- Counts files in a temporary directory
- Pushes metrics to Prometheus Pushgateway
- Enables visualization in Grafana

- Supports alerting based on threshold values

Containerization ensures:

Consistent runtime
Dependency portability
Reusable deployment across multiple AWS services
Same application behavior across environments

The same container image was reused for both Lambda and ECS deployments.

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3. Containerization in AWS Lambda

3.1 Overview

AWS Lambda supports execution of container images instead of traditional ZIP-based functions.

In this approach:

- The monitoring collector is packaged as a Docker image.
- The image is stored in Amazon ECR.
- Lambda executes the container only when triggered.

3.2 Execution Model

Lambda follows a **serverless event-driven execution model**.

Characteristics:

- Container runs only when invoked.
- Automatically scales based on events.
- No server or cluster management required.
- Execution is short-lived.

In this task:

- Lambda was triggered using **EventBridge scheduler** every few minutes.
- The container executed the collector logic.
- Metric was pushed to Pushgateway.
- Execution terminated automatically.

3.3 Key Features

- Fully managed infrastructure
- Automatic scaling
- Pay-per-execution pricing
- Stateless execution
- Minimal operational overhead

3.4 Use Case Suitability

Lambda container deployment is ideal when:

- Tasks are periodic or event-driven
- Execution duration is short
- Infrastructure management should be minimized

Example from task:

> Scheduled temp directory monitoring every 5 minutes.

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4. Containerization in AWS ECS Fargate

4.1 Overview

Amazon ECS Fargate runs containers without managing servers while still providing container orchestration capabilities.

Here:

- The same monitoring container image is deployed as an ECS task.
- Fargate provisions compute resources automatically.

4.2 Execution Model

ECS follows a **container orchestration model**.

Characteristics:

- Containers run as tasks or services.
- Execution duration can be long-running.
- Scheduling controlled via ECS or EventBridge.

- More runtime control compared to Lambda.

In this task:

- The collector container runs as a scheduled ECS task.
- Pushes metrics periodically.
- Logs stored in CloudWatch.

4.3 Key Features

- Long-running container support
- Custom CPU and memory allocation
- Networking flexibility
- Persistent execution capability
- Better suited for continuous workloads

4.4 Use Case Suitability

ECS Fargate deployment is ideal when:

- Continuous monitoring is required
- Longer execution time is needed
- More control over container resources is required

Example from task:

> Continuous monitoring collector running independently of Lambda limits.

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5. Key Differences Between Lambda and ECS Containerization

Feature	Lambda Container	ECS Fargate Container
Execution Type	Event-driven	Task / Service based
Infrastructure Management	Fully serverless	Managed container orchestration
Startup	On-demand	Always running or scheduled
Scaling	Automatic per invocation	Manual / Auto scaling
Execution Duration	Limited	Long-running supported

Cost Model	Pay per execution	Pay for running resources
Operational Control	Minimal	Higher control
Best For	Scheduled lightweight jobs	Continuous workloads

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6. Architectural Comparison in This Project

Lambda Deployment Flow

```
EventBridge Scheduler
        ↓
Lambda Container
        ↓
Temp Directory Scan
        ↓
Pushgateway
        ↓
Grafana Dashboard
```

ECS Deployment Flow

```
EventBridge / ECS Scheduler
        ↓
ECS Fargate Task
        ↓
Collector Container
        ↓
Pushgateway
        ↓
Grafana Dashboard
```

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7. Advantages Observed in This Task

Lambda Advantages

- Faster deployment
- Zero infrastructure management
- Ideal for periodic monitoring

ECS Advantages

- Better for persistent collectors
- Greater runtime flexibility
- Resource customization

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8. Conclusion

Both AWS Lambda and ECS Fargate enable containerized execution but serve different operational needs.

In this monitoring solution:

- **Lambda containerization** provided a lightweight serverless execution model suitable for scheduled metric collection.
- **ECS containerization** enabled scalable and controlled container execution for extended monitoring scenarios.

Using both approaches demonstrates deployment portability and highlights how the same containerized application can operate across different AWS compute environments.